National University of Computer & Emerging Sciences, Karachi
FAST School of Computing
Midterm 2 Examination, Spring 2023
3rd April 2023, 11:30 am – 12:30 pm

Course Code: CS3001	Course Name: Computer Networks
Instructor Names: Dr. Sufian Hameed, Dr. Farrukh Salim, Mr. Shoaib Raza, and Ms. Yusra Kaleem	
Student Roll No:	Section No:

Instructions:

- Return the question paper.
- **All questions must be answered in answer script and according to the sequence given in the question paper.**
- In case of any ambiguity, you may make an assumption. But your assumption should not contradict any statement in the question paper.

Time: 60 minutes. **Max Marks:** 50 Points

Question 1: **[CLO-2]** **[10 points]**

Recall reliable data transfer over a lossy channel with bit errors: rdt 3.0. In case of packet losses the protocol uses a time based retransmission mechanism which requires a countdown timer which interrupts the sender after a given timeout. Hence in such a protocol the sender will need be able to **(a)** start the timer each time a packet (either a first-time packet or a retransmission) is sent **(b)** respond to a timer interrupt (taking appropriate actions) **(c)** stop the timer. Given the FSM (Finite State Machine) of a sender in the figure 1 please provide the FSM (Finite State Machine) of the receiver.

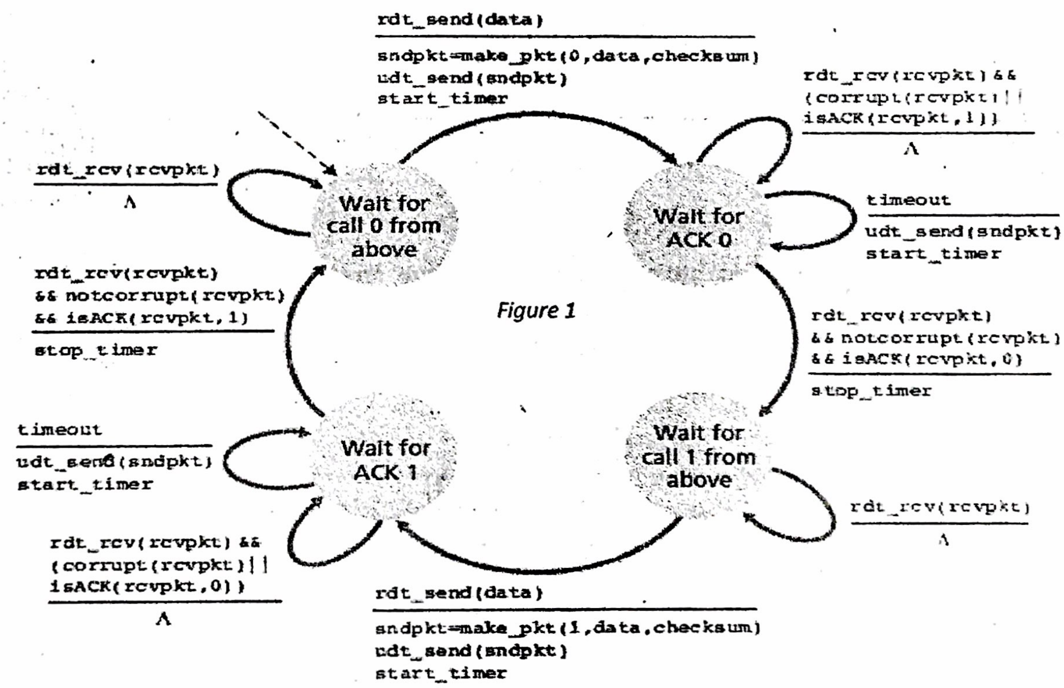


Figure 1

Question 2: **[CLO-1]** **[10 points]**

The network shown in figure 2 perform two parallel TCP transmissions. TCP1 is a transmission between Source A and Sink A that uses TCP Tahoe. TCP2 is a transmission between Source B and Sink B that uses TCP Reno. Initial ssthresh for both TCP transmissions is set to 32. In this specific scenario no additional delay through forwarding is introduced. Thus, the RTT is only composed of the sums of the delay indicated on each link, times two.

(a) For the TCP 1 transmission, draw the resulting congestion window, assuming that a packet loss (triple duplicate ACKs) is detected at time t=900ms.

(b) For the TCP 2 transmission, draw the resulting congestion window, assuming that a packet loss (triple duplicate ACKs) is detected at time t=650ms.

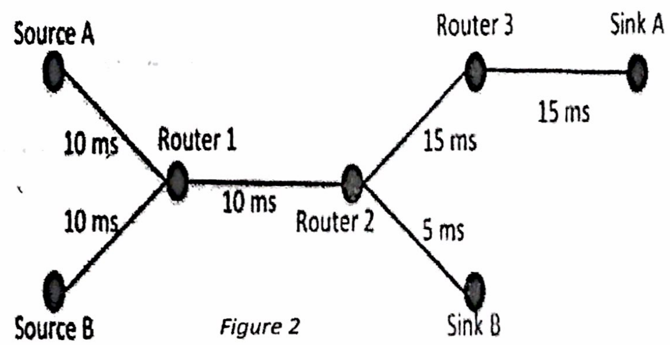


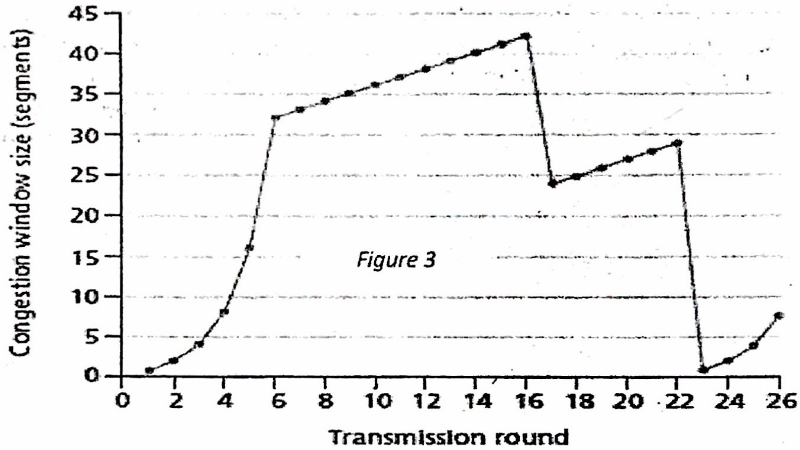
Figure 2

Question 3: **[CLO-1]** **[10 points]**

Draw a time-sequence diagram for the Selective Repeat algorithm with the sender's window size equal to 4. Assume at t=0, the sender sends packets 0, 1, 2, 3 and packet 2 will be lost.

Question 4: **[CLO-3]** **[10 points]**

Assuming TCP Reno is experiencing the behavior shown in the following figure 3.



Answer the following questions. In all cases, you should provide a short discussion justifying your answer.

(a) Identify the intervals of time when TCP slow start is operating.
(b) Identify the intervals of time when TCP congestion avoidance is operating.
(c) After the 16th transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?
(d) After the 22nd transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?
(e) What is the initial value of ssthresh at the first transmission round?
(f)What is the value of ssthresh at the 18th transmission round?
(g) What is the value of ssthresh at the 24th transmission round?
(h) During what transmission round is the 70th segment sent?
(i) Assuming a packet loss is detected after the 26th round by the receipt of a triple duplicate ACK, what will be the values of the congestion window size and of ssthresh?

Question 5: **[CLO-2]** **[10 points]**

NUCES has been assigned the network **12.56.1.0/24** and wants to divide it among the five Campuses. **Lahore** needs to accommodate up to 126 hosts, **Karachi** needs to accommodate up to 62 hosts, **Islamabad** needs to accommodate up to 30 hosts, **Peshawar** needs to accommodate up to 14 hosts and **Faisalabad** needs to accommodate up to 6 hosts. Fill the following table 01 in your answer script with the details of the sub-networks that the NUCES can create to fit its campuses' needs.

Subnet No.	Network Address	Netmask	Host Range	No. Of Hosts
Lahore				
Karachi				
Islamabad				
Peshawar				
Faisalabad				

Table 01

Best of Luck!!!